

MnemoMaker: Creator, Curator, or Something Else? Exploring Human-AI Mnemonic Co-Creation

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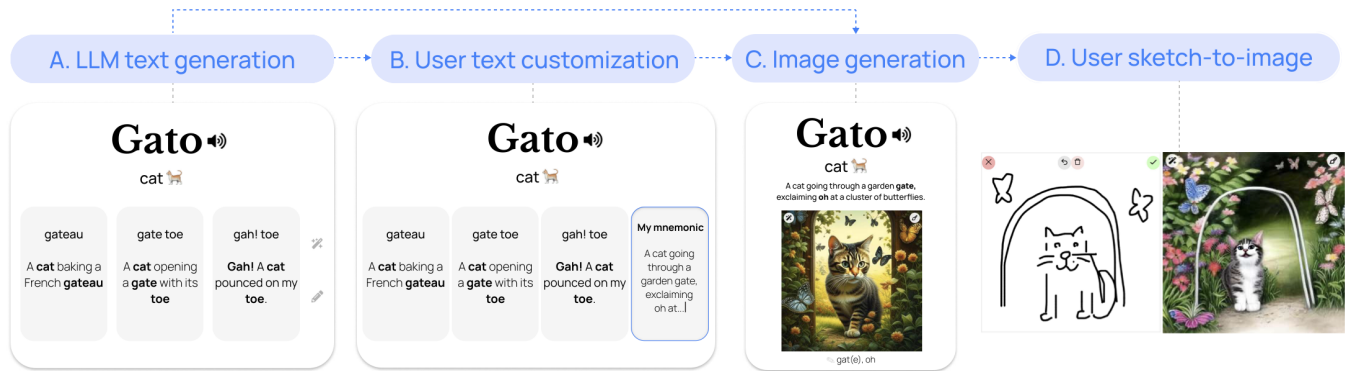


Figure 1: MnemoMaker's co-creation workflow for generating and adapting AI-generated verbal and visual mnemonics.

Abstract

MnemoMaker is an interactive system enabling learners to co-create personalized verbal and visual mnemonics with AI assistance. Through iterative design with 87 participants across three phases, we found that likeability strongly predicts perceived memorability ($R^2 = .588, p < .001$), with effective mnemonics characterized as creative, imaginable, and emotionally engaging. Users reported strong ownership over AI-assisted creations (5.20/7) despite attributing them nearly equally to themselves and the system (3.54 on a 1-7 scale from "system's contributions" to "my contributions"). *How* people engage with AI, not just *what* they create, shapes their sense of agency. Our demo invites CHI attendees to create vocabulary mnemonics in any language and discover their collaboration style (e.g., as Creators, Curators, Negotiators, Acceptors, or something in

between). A live dashboard visualizes collaboration patterns across all attendees, turning individual experiences into collective insight about human-AI co-creation.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; **Empirical studies in HCI**; *Interactive systems and tools*.

Keywords

AI co-creation, human-AI collaboration, language learning, vocabulary learning, mnemonics, generative AI

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1 Introduction

Mnemonics are memory aids that create meaningful associations between new information and existing knowledge. The keyword method, which links foreign words to similar-sounding native words through mental imagery, has been validated for vocabulary learning [3, 5, 6]. However, creating personalized mnemonics requires significant cognitive effort, and the technique remains underutilized [8].

Prior systems have explored automated mnemonic generation. TransPhoner [10] uses algorithmic phonetic matching but produces no visuals, while SmartPhone [4] combines LLM-generated verbal mnemonics with images but offers no user customization. In both cases, learners are passive recipients, which risks undermining the *generation effect* (i.e., self-generated content is remembered better than passively received information [12]), particularly when active, multimodal encoding is involved [1, 13]. Complementary work has explored generative environments for contextual vocabulary practice [11]. Research on human-AI collaboration suggests that users prefer to lead while AI assists [7] and that bidirectional communication enhances the sense of partnership [9]. We call this balance *scaffolded agency*.

We present MnemoMaker, an interactive system where learners *co-create* personalized mnemonics with AI. Rather than fully automating generation, MnemoMaker scaffolds a workflow where users can accept, modify, or replace AI suggestions. Users can also draw their own visual mnemonics and enhance them with AI assistance.

Our contributions are two-fold. First, through iterative design with 87 participants across three phases, we reveal an *ownership paradox* in human-AI co-creation where users feel strong ownership over AI-assisted creations despite attributing them nearly equally to themselves and the system. The *how* of engagement matters as much as the *what* of creation. Second, we present an interactive demo where CHI attendees create personalized mnemonics in any language and discover their collaboration style. A live dashboard visualizes patterns across attendees, transforming individual experiences into collective insight about human-AI co-creation.

2 Iterative Design Process

MnemoMaker emerged through a human-centered design process involving 87 participants across three phases, with IRB approval. A preliminary survey (n=38) revealed that 71% of language learners cited time constraints as the primary barrier to creating mnemonics, 84% preferred combined visual and verbal mnemonics, and 42% reported abandoning mnemonic creation when stuck for ideas. These findings motivated our core design goal of reducing effort while preserving personalization and multimodal encoding.

A one-week beta deployment (n=14) led to critical refinements including reducing default mnemonic complexity, adding progressive disclosure of advanced features, and improving visual coherence. In our final exploratory study (n=35), participants each created mnemonics for 16 vocabulary words (eight Spanish, eight advanced English). We collected likeability ratings, perceived memorability ratings, and system usability metrics. Figure 2 shows example cards created during the study.

3 System Design

MnemoMaker supports a co-creation workflow designed to accommodate diverse collaboration approaches (Figure 1). Figure 3 shows the system architecture.

3.1 Verbal Mnemonic Generation

When users enter a vocabulary word, MnemoMaker generates multiple verbal mnemonic suggestions using GPT-4o (Figure 4). The system creates phonetic associations linking foreign words to familiar sounds. For example, the Spanish word “perro” (dog) might yield “A **pair** of *dogs* **rowing** a boat.” Each mnemonic is stylized for readability, with bold text highlighting the homophone and italics indicating the definition. It uses GPT-4o-mini as a content moderator to validate input vocabulary and flag inappropriate terms before mnemonic generation.

Prior work noted that generating descriptive cues for abstract words is challenging [4]. We mitigate this by exposing the model to examples that ground abstract concepts in visual symbolisms, such as transforming “gracias” (thank you) into “two **grassy** hands grasping a heart, symbolizing *thankfulness* for **us**.” Users can browse multiple suggestions, regenerate alternatives, or create their own mnemonic from scratch (Figure 5).

3.2 Visual Mnemonic Generation

Selected verbal mnemonics are transformed into visual representations using FLUX Schnell. Users can choose from, or randomize, six visual styles to match their preferences (Figure 6). Prior work using DALL-E 2 found it struggled to combine disparate concepts [4]. MnemoMaker uses a ControlNet-based architecture [15] that produces more spatially coherent outputs. Users can regenerate images or adjust the input text to guide results.

3.3 Sketch Enhancement

MnemoMaker also provides a sketch canvas where user drawings can be enhanced via Scribble Diffusion, maintaining the user’s composition while adding detail (Figure 6, right).

3.4 Technical Implementation

MnemoMaker is built with React and Node.js/Express, using Firebase for data persistence. AI services include OpenAI GPT-4o for text and Replicate for image generation.

4 Key Findings

Our exploratory study with 538 created cards revealed insights relevant to human-AI co-creation in learning contexts.

Likeability predicts memorability. Participants rated mnemonics as generally likeable ($M=5.17/7$) and memorable ($M=5.79/7$), with likeability strongly predicting perceived memorability ($R^2=.588$, $p<.001$). This aligns with the *mnemonic effect of choice* (i.e., learners remember content better when they have agency in selecting it [2]). Liked cards succeeded when they were creative (26 participants), fit well with the word’s meaning (23), and were imaginable as coherent mental scenes (18). As P17 put it, “Most cards aptly convey their meanings and exude an artistic, memorable quality.”

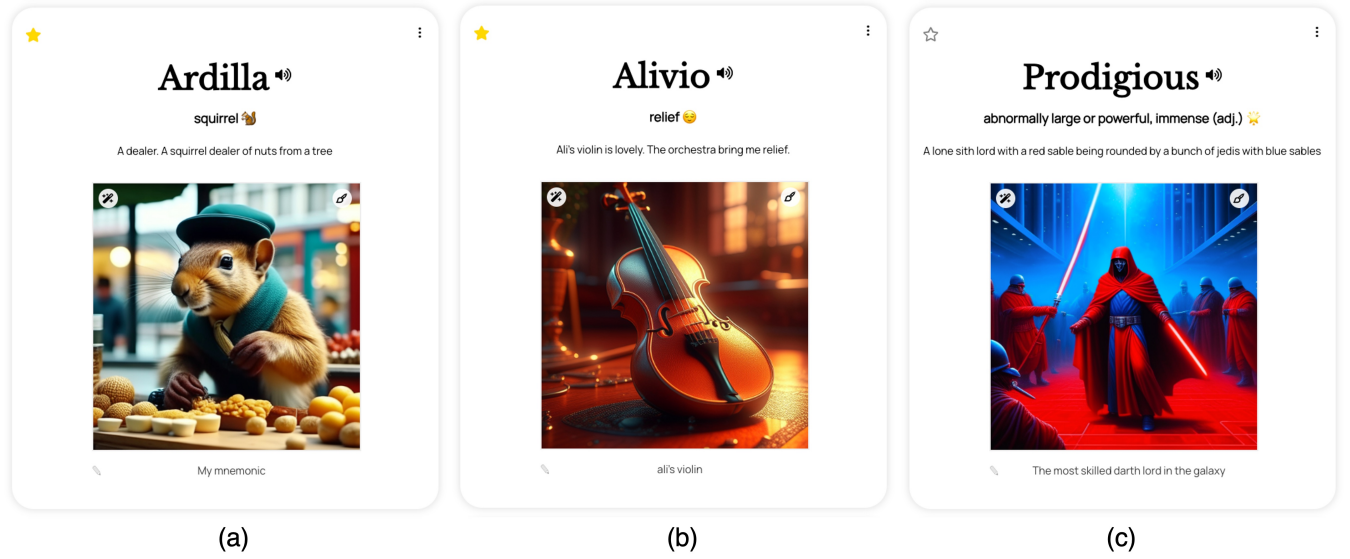


Figure 2: Example mnemonic cards showing word, definition, verbal and visual mnemonics (e.g., (a) “ardilla”/squirrel → “a dealer”, depicting an anthropomorphic squirrel vendor dealing nuts)

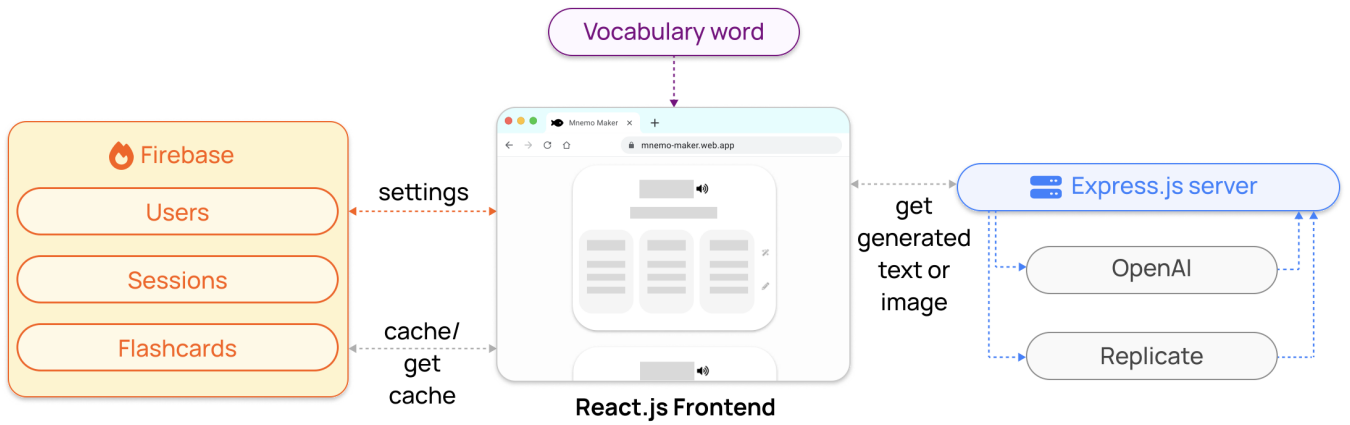


Figure 3: System architecture of MnemoMaker.

Table 1: System evaluation results (7-point scale)

Metric	Mean (SD)
Easy to use	6.46 (0.85)
Engaging	6.11 (0.96)
Useful for learning	6.09 (1.07)
Future use intent	6.09 (1.25)
Felt like collaboration	5.77 (1.37)
Sense of ownership	5.20 (1.69)

Disliked cards failed primarily due to poor semantic fit (16 participants) or incoherence (13). Most strikingly, 15 participants described disliked cards as “strange,” a term that emerged specifically

for AI-generated artifacts. The squirrel dealer mnemonic for “ardilla” (Figure 2a) exemplifies productive creative bizarreness, feeling unexpected but coherent. What participants rejected were outputs that missed semantic connections or produced uncanny compositions.

The ownership paradox. Users reported strong ownership over AI-assisted creations (5.20/7), yet attributed them nearly equally to themselves and the system (3.54 on a 1-7 scale from “system’s contributions” to “my contributions”). Ownership in human-AI collaboration depends less on *who did the work* and more on *how the collaboration felt*. Consistent with design principles for generative AI that emphasize user agency in co-creation [14], we found that the ability to browse, select, and modify appears sufficient to generate ownership, even when most users accepted AI suggestions without

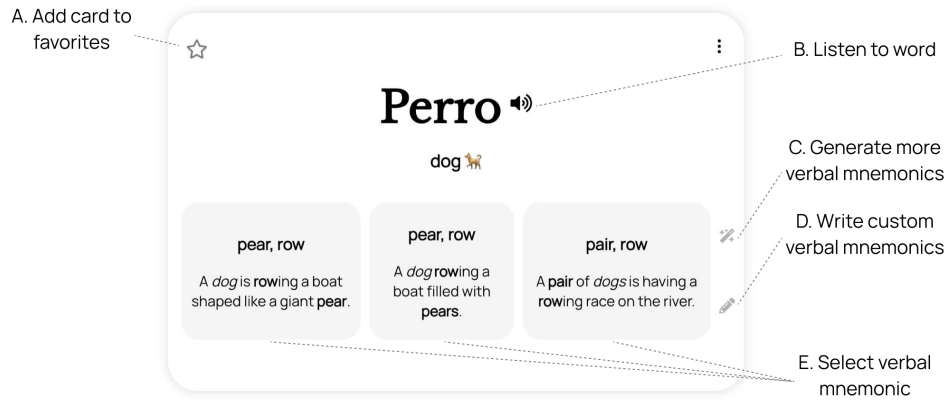


Figure 4: Card interface for a vocabulary word. New verbal mnemonic suggestions are added three at a time via pagination.

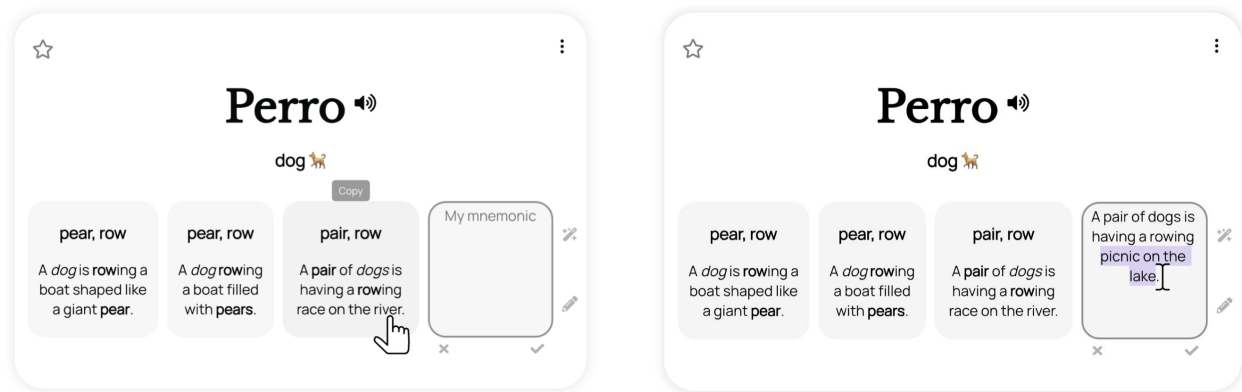


Figure 5: Process for creating custom verbal mnemonics: (left) optionally copying an existing suggestion as a starting point, (right) entering a custom mnemonic in the text field.

modification. Our demo’s “See My Style” feature emerged from this insight.

System reception. Participants rated the system highly across multiple dimensions (Table 1). Ratings for collaboration (5.77/7) and ownership (5.20/7) indicate users perceived the experience as genuinely collaborative despite AI generating the initial content.

5 Demo Experience

Our demo invites CHI attendees to explore how they co-create with AI, accessible at a URL shared at CHI. Attendees can try any language pair they know, though the phonetic bridging for mnemonics works best when languages share sounds (e.g., Spanish-English yields closer matches than Arabic-Chinese).

Upon entering the demo, attendees see live community statistics displaying the number of cards created and the distribution of collaboration styles among previous users. They choose a username and select their native and target languages. After an optional guided example, attendees create mnemonic cards by typing in vocabulary words or using the random word generator to quickly

surface options. For each word, they browse AI-generated verbal suggestions, select or modify one, and generate a visual mnemonic in their preferred style. The sketch canvas lets users draw their own visuals and enhance them with AI. These interactions shape a personalized collaboration profile for each attendee.

Clicking “See My Style” reveals the personalized collaboration profile. **Creators** prefer generating original content through sketches and custom mnemonics. **Curators** browse many options before selecting, spending time evaluating alternatives. **Negotiators** frequently modify and refine AI suggestions, shaping outputs to match their vision. **Acceptors** efficiently accept suggestions that work, prioritizing speed over extensive customization. These labels, including in between, give attendees vocabulary to reflect on their collaboration tendencies and compare approaches with others.

All attendee-generated cards appear in a shared gallery alongside a live dashboard displaying the distribution of collaboration styles across CHI attendees. The gallery updates in real time as new cards are created, filterable by target language. Attendees can favorite mnemonics and browse community favorites. This creates a social

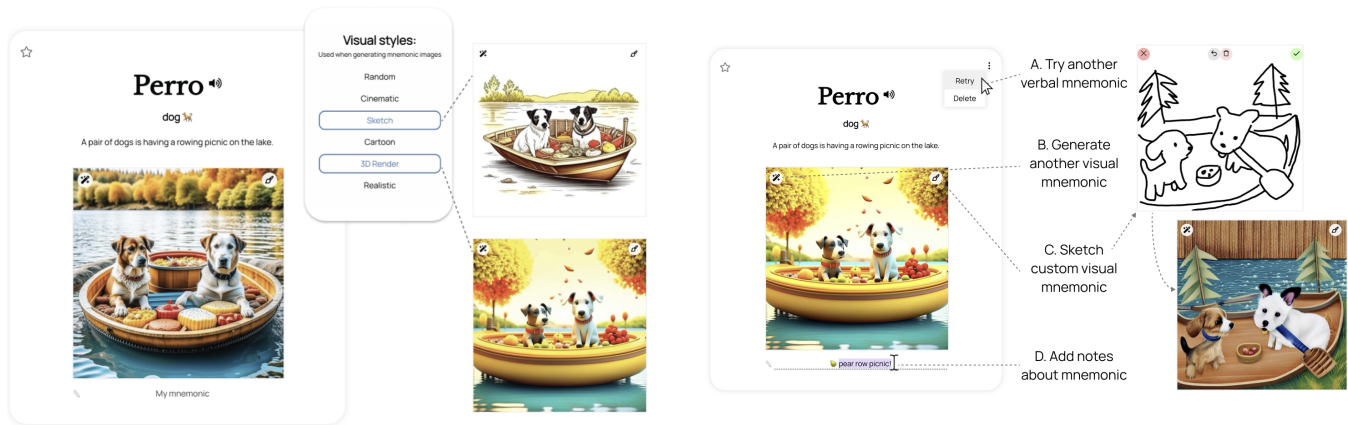


Figure 6: Left: Six visual style options (Random, Cinematic, Sketch, Cartoon, 3D Render, Realistic) with examples showing same mnemonic rendered in different styles. Right: Card menu for retrying verbal mnemonics or deleting cards, plus sketch canvas with AI enhancement on user-input line drawings.

dimension where individual experiences aggregate into collective insight about how the CHI community engages with generative AI.

6 Discussion and Future Work

The tension between ownership and attribution hints at something fundamental about learning in the age of generative AI. When a student feels ownership over something the system largely created, is that productive self-deception or a legitimate path to engagement? Our findings suggest that meaningful choice matters more than effort expended, and future work should validate this through longitudinal retention studies.

The collaboration styles we identified remain descriptive rather than predictive. More rigorous taxonomies could help researchers understand individual differences in how people work with generative systems. Our ongoing goal is to identify the minimum viable effort required to preserve learning benefits while maximizing accessibility.

7 Conclusion

MnemoMaker explores AI-mediated learning where the system generates possibilities but learners retain choice. Attendees will create mnemonics in their own languages, discover whether they create, curate, negotiate, or accept, and see their patterns reflected in the collective behavior of the CHI community.

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